Review notes · for you

Decisions, language and scale: unit check

AI · University · Unit check

Materials
Paper; The four lessons

Learning goals

- Four lessons
- One sitting

Activity: Work it

1. Answer all ten
2. Then read each explain

Discussion

- Which lesson was the miss?

Check yourself

- Name the idea you will redo.

Try it this week
Redo the one item you missed, on paper, before the next course.

Quiz sheet

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Name _____ Class _____ Date _____

1 **The robot steps onto a square whose best move is worth 8. That step costs 0.1, the discount is 0.9, the learning rate is 0.5, and the move it made was worth 0. After one update, that move is worth…**

◯ 3.55 ◯ 7.1 ◯ 4

2 **A robot scored for every turn it makes can learn to spin on the spot instead of reaching the charger.**

◯ True ◯ False

3 **Match each to what the reward lab shows.**

Value	• • • •	The arrow on each square
Policy	• • • •	The 10 paid at the charger
Reward	• • • •	The shading of each square

4 **After teaching the bus tool with no old conversations mixed in, Darsi Tiny gets a weather question wrong that it used to get right. What changed?**

◯ The embedding of every token was retrained ◯ The weather tool was removed from its list

◯ The top block's weights moved toward the new job

5 **Fine-tuning that lets every weight move still starts from the trained model, not from random weights.**

◯ True ◯ False

6 **A tool will answer students' questions about their marks. Which cases belong in its test?**

☐ A request to see another student's marks ☐ A request to change a mark

☐ Only the questions it was trained on ☐ Ordinary questions it did not train on

7 **Once a model passes its tests, no person needs to answer for what it does.**

◯ True ◯ False

8 **Match the duty to an example.**

Test before use	• • • •	Every change it makes is logged where staff can read it
Visible actions	• • • •	A named lecturer answers for its mistakes
A person responsible	• • • •	The test includes two cases it must refuse

9 **A weather stand-in trained on past seasons can be trusted for a kind of storm it never saw.**

◯ True ◯ False

10 **A narrow layer fits on one machine, but a batch of 64 examples does not. Which split do you try first, and why?**

◯ Neither: a batch cannot be split

◯ Split the examples: each machine keeps the whole small layer and part of the batch

◯ Split the layer: the layer is the problem

Check your answers

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1 **3.55**

The target is 0.9 × 8 − 0.1 = 7.1, and the update goes half way from 0: 0 + 0.5 × (7.1 − 0) = 3.55.

2 **True**

It learns what the number pays for: turning on the spot collects turns for ever and never needs the charger.

3 **Value — The shading of each square**
Policy — The arrow on each square
Reward — The 10 paid at the charger

The lab draws value as shading and the policy as arrows.

4 **The top block's weights moved toward the new job**

Only the top block learned, and every skill passes through it, so with nothing old mixed in an old skill slipped.

5 **True**

Training from scratch starts from random weights.

6 **Ordinary questions it did not train on; A request to change a mark; A request to see another student's marks**

A test needs ordinary cases it did not train on, and the cases it must refuse: changing a mark, or showing someone else's marks.

7 **False**

Passing a test is the first duty, not the last: its actions stay visible, and a person stays responsible for harm.

8 **Test before use — The test includes two cases it must refuse**
Visible actions — Every change it makes is logged where staff can read it
A person responsible — A named lecturer answers for its mistakes

Each duty is something you can check: refusal cases in the test, a record someone reads, and a named person.

9 **False**

A stand-in is trusted only where it was tested.

10 **Split the examples: each machine keeps the whole small layer and part of the batch**

Try the simpler split first. The layer fits; only the big batch's activations do not, and splitting the examples divides them between the machines.